

STUDENTHUBMOBILEAPPLICATION.

NAME: CELESTINE GITAU

REGISTRATION NUMBER: BIT/2024/54146

COURSE: BIT

UNIT: MOBILE APPLICATION DEVELOPMENT

PROJECT: STUDENTHUB APPLICATION

1. INTRODUCTION

Student Hub is an Android mobile application developed using Java, XML, and SQLite Database in Android Studio. The application enables students to register accounts, log into the system, access the dashboard, view profile details, and check fee structure information through a mobile interface.

The application was designed and tested successfully using the Android Emulator.

WEEK 1: PROJECT SETUP & USER INTERFACE DESIGN

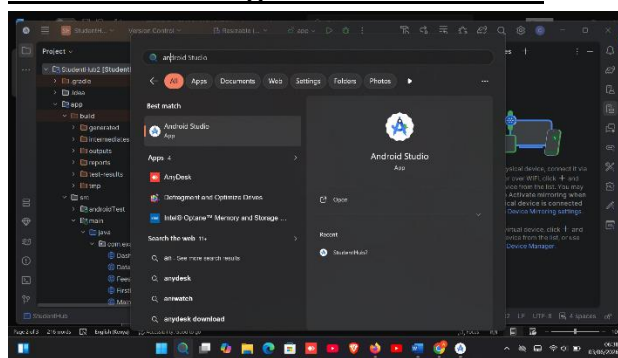
Objectives

- Install and configure Android Studio
- Create Student Hub project
- Set up Android Emulator
- Understand Android project structure
- Design Login and Registration interfaces

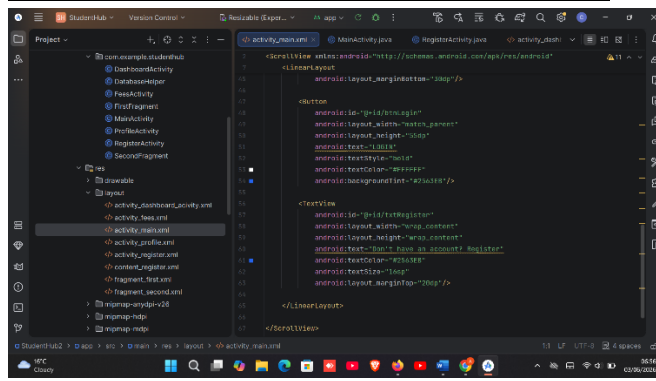
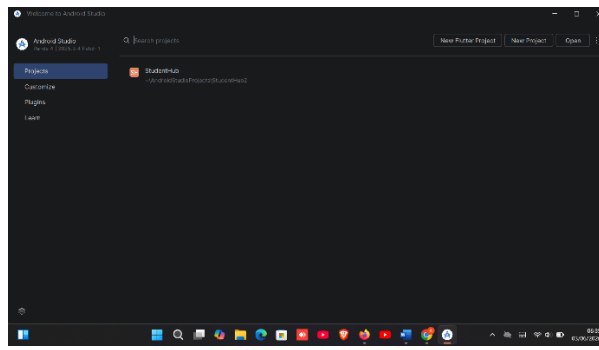
Its built

I spent this week setting up the working environment. For the interface, I used a flexible layout system in XML to position the login boxes, text labels, and click buttons. This ensures that the text boxes and forms align nicely and stay in the center of the screen, no matter how big or small the phone screen is.

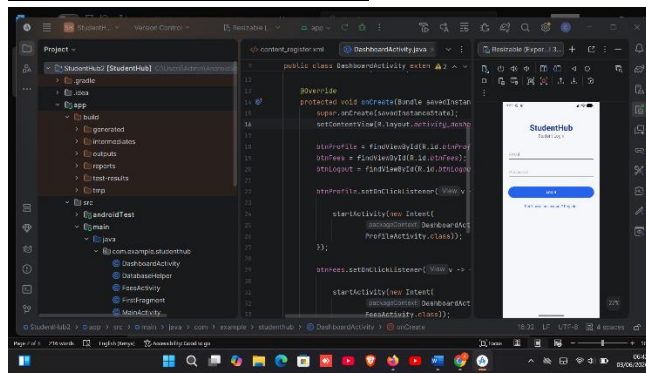
1. Install and configure android studio



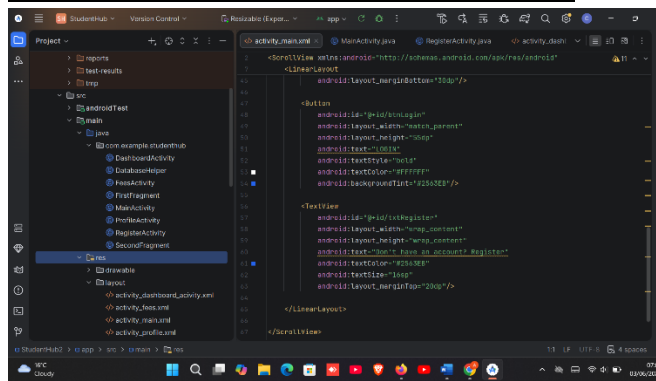
2. Create student hub project



3. Setup android emulator



4. Understand android project structure



5. Design logic and instruction interface

Activity login.xml.

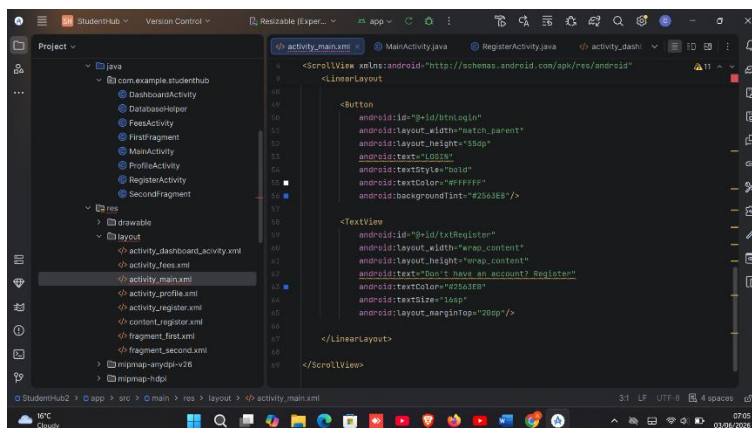
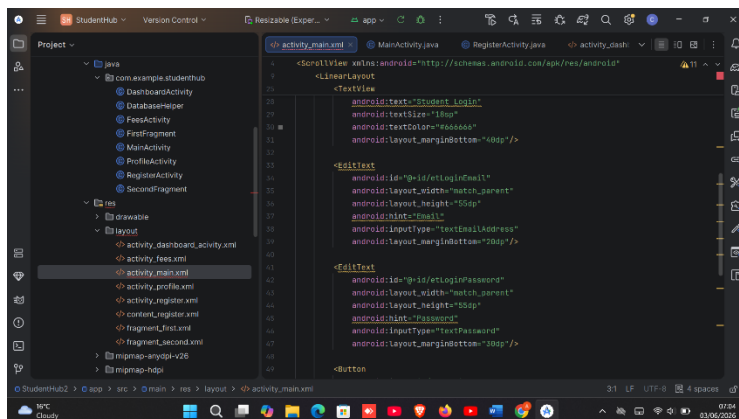
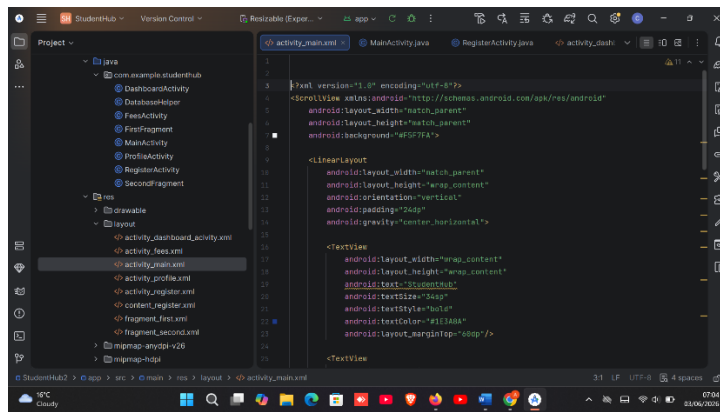
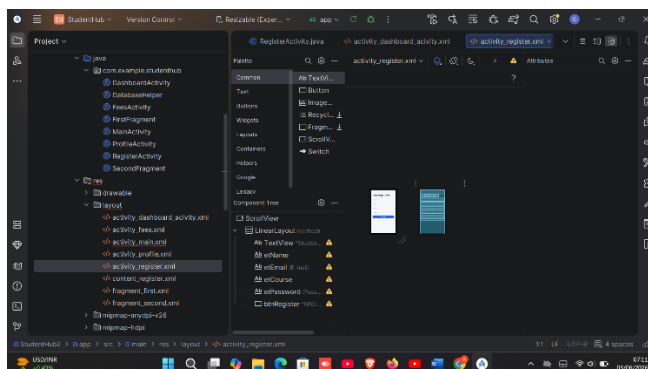
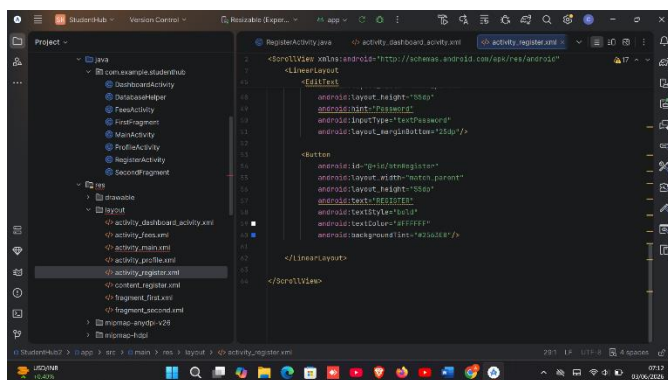
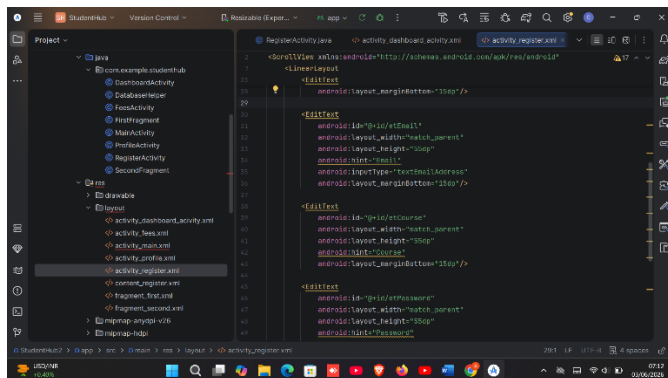
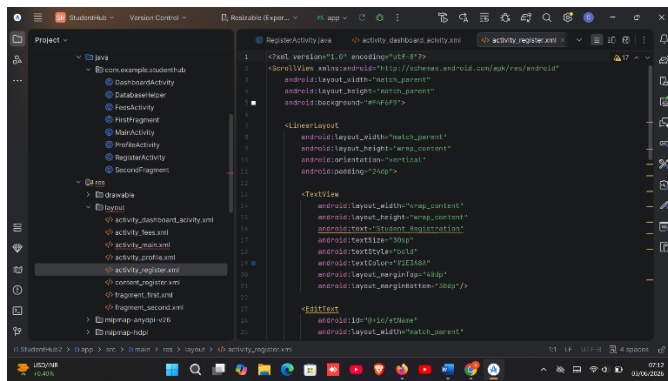


Figure 1. code view of activity_login.xml

Activity register.xml





WEEK 2: LOGIN AND REGISTRATION FUNCTIONALITY

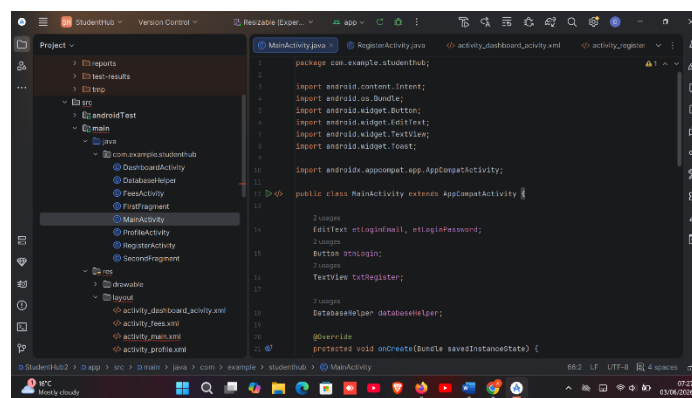
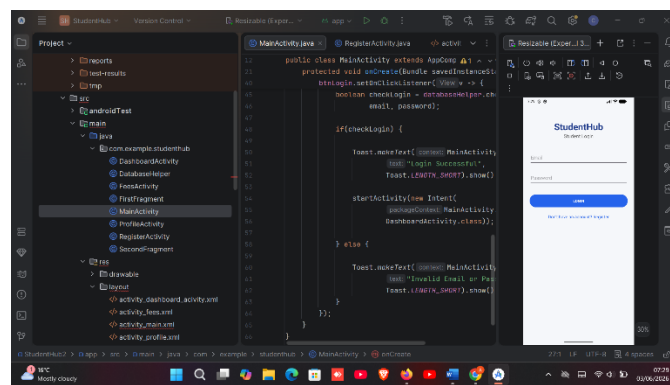
Objectives

- Implement login functionality
- Implement registration functionality
- Validate user credentials
- Navigate between activities
- Create Dashboard screen

Explanation on its built

I connected the XML user interface screens to the Java code by linking text fields and buttons using their IDs. I then added input validation to ensure users enter the required information correctly. For example, the app displays warning messages when fields are left empty or when a password is too short. I also implemented button click events that allow users to move from the login screen to the main dashboard after entering valid details.

i. Implement log in functionality



```

11 public class MainActivity extends AppCompatActivity {
12     protected void onCreate(Bundle savedInstanceState) {
13         setContentView(R.layout.activity_main);
14
15         etLoginEmail = findViewById(R.id.etLoginEmail);
16         etLoginPassword = findViewById(R.id.etLoginPassword);
17
18         btnLogin = findViewById(R.id.btnLogin);
19         txtRegister = findViewById(R.id.txtRegister);
20
21         databaseHelper = new DatabaseHelper(this);
22
23         txtRegister.setOnClickListener(View v -> {
24             startActivity(new Intent(
25                 packageContext, RegisterActivity.class));
26         });
27
28         btnLogin.setOnClickListener(View v -> {
29             String email = etLoginEmail.getText().toString();
30             String password = etLoginPassword.getText().toString();
31
32             boolean checkLogin = databaseHelper.checkLogin(
33                 email, password);
34         });
35     }
36 }

```

```

11 public class MainActivity extends AppCompatActivity {
12     protected void onCreate(Bundle savedInstanceState) {
13         setContentView(R.layout.activity_main);
14
15         etLoginEmail = findViewById(R.id.etLoginEmail);
16         etLoginPassword = findViewById(R.id.etLoginPassword);
17
18         btnLogin = findViewById(R.id.btnLogin);
19         txtRegister = findViewById(R.id.txtRegister);
20
21         databaseHelper = new DatabaseHelper(this);
22
23         txtRegister.setOnClickListener(View v -> {
24             startActivity(new Intent(
25                 packageContext, RegisterActivity.class));
26         });
27
28         btnLogin.setOnClickListener(View v -> {
29             String email = etLoginEmail.getText().toString();
30             String password = etLoginPassword.getText().toString();
31
32             boolean checkLogin = databaseHelper.checkLogin(
33                 email, password);
34
35             if (checkLogin) {
36                 Toast.makeText(this, "Login Successful",
37                     Toast.LENGTH_SHORT).show();
38             } else {
39                 Toast.makeText(this, "Invalid Email or Password",
40                     Toast.LENGTH_SHORT).show();
41             }
42         });
43     }
44 }

```

Figure 2mainactivity.java

ii. Implement registration functionality

```

1 <?xml version="1.0" encoding="utf-8"?>
2 <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
3     android:layout_width="match_parent"
4     android:layout_height="match_parent"
5     android:orientation="vertical"
6     android:padding="16dp"
7     >
8     <TextView
9         android:id="@+id/tvTitle"
10        android:layout_width="match_parent"
11        android:layout_height="wrap_content"
12        android:text="Student Registration"
13        android:textSize="24sp"
14        android:gravity="center"
15        >
16
17     <EditText
18         android:id="@+id/etName"
19         android:layout_width="match_parent"
20         android:layout_height="wrap_content"
21         android:hint="Name"
22         >
23
24     <EditText
25         android:id="@+id/etEmail"
26         android:layout_width="match_parent"
27         android:layout_height="wrap_content"
28         android:hint="Email"
29         >
30
31     <EditText
32         android:id="@+id/etPassword"
33         android:layout_width="match_parent"
34         android:layout_height="wrap_content"
35         android:hint="Password"
36         android:inputType="password"
37         >
38
39     <Button
40         android:id="@+id/btnRegister"
41         android:layout_width="match_parent"
42         android:layout_height="wrap_content"
43         android:text="Register"
44         >
45
46 </LinearLayout>

```

```

1 package com.example.studenthub;
2
3 import android.content.Intent;
4 import android.os.Bundle;
5 import android.widget.Button;
6 import android.widget.EditText;
7 import android.widget.Toast;
8
9 import androidx.appcompat.app.AppCompatActivity;
10
11 public class RegisterActivity extends AppCompatActivity {
12
13     EditText etName, etEmail, etPassword;
14     Button btnRegister;
15
16     DatabaseHelper databaseHelper;
17
18     @Override
19     protected void onCreate(Bundle savedInstanceState) {
20         super.onCreate(savedInstanceState);
21         setContentView(R.layout.activity_register);
22     }
23 }

```

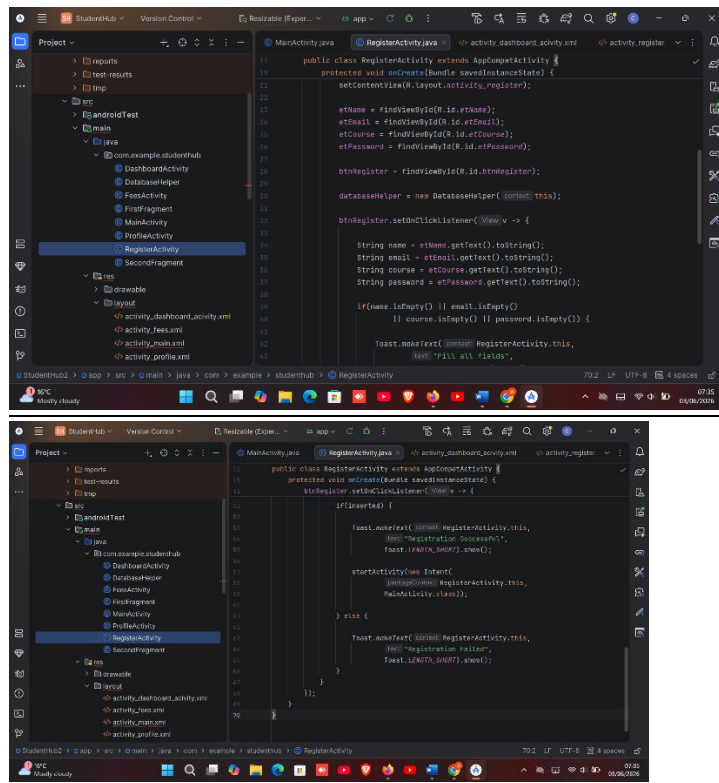
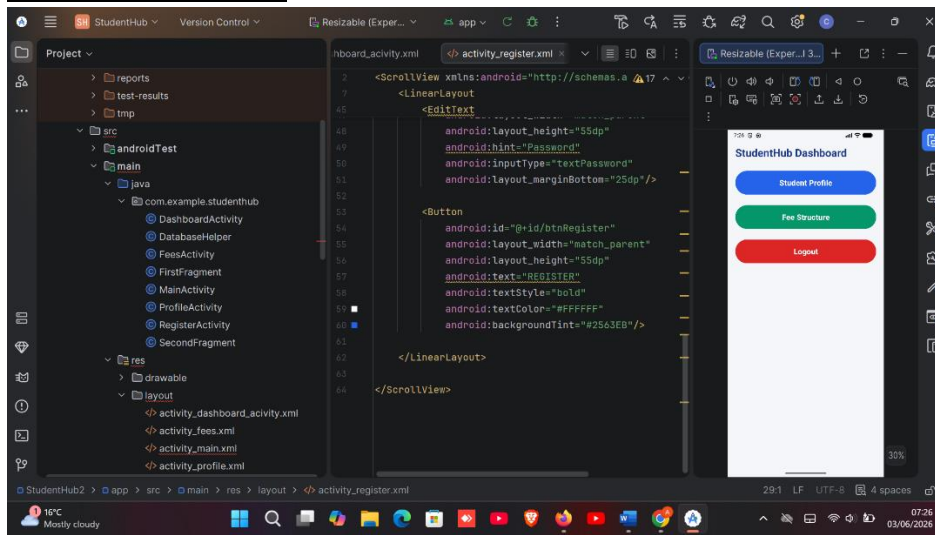


Figure 3 registeractivity.java

DASHBOARD SCREENS



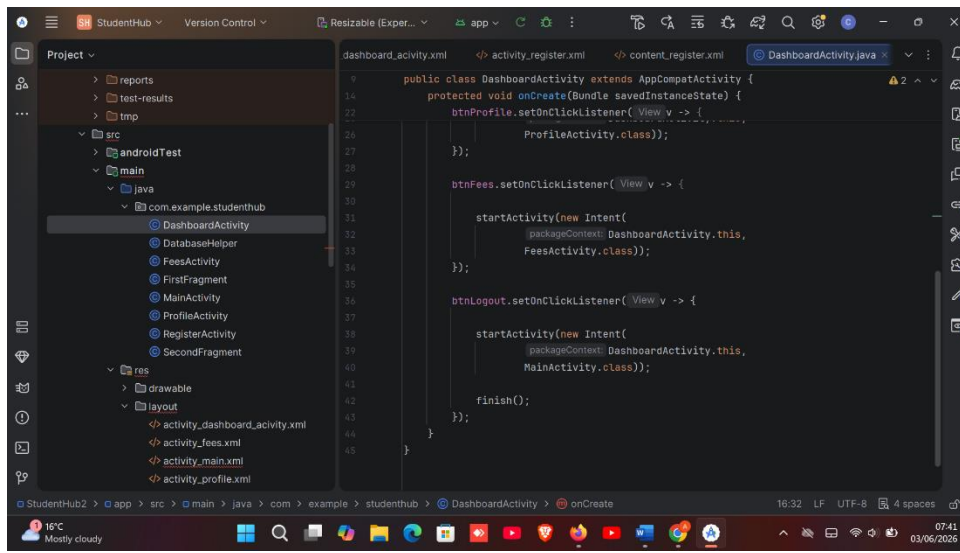
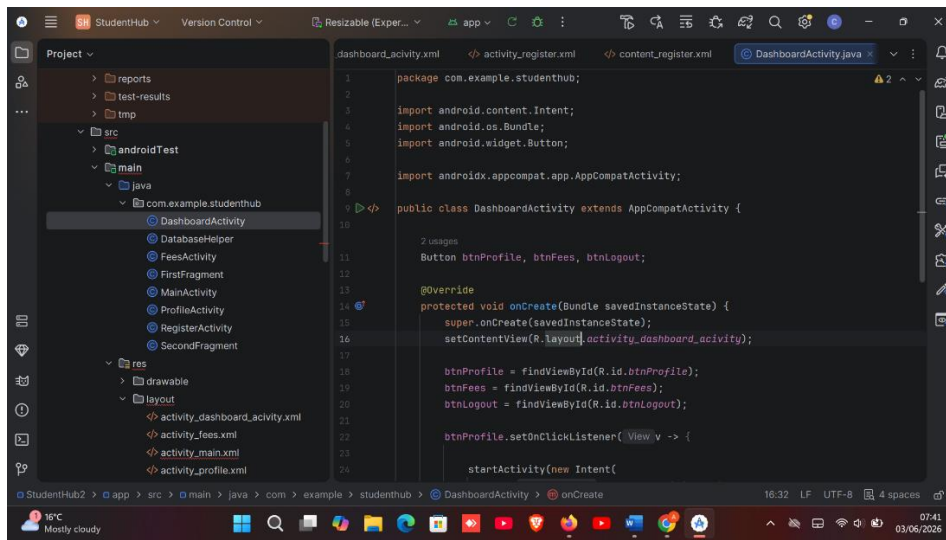


Figure 4. Dashboardactivity.java

4. WEEK 3: SQLITE DATABASE & APPLICATION NAVIGATION

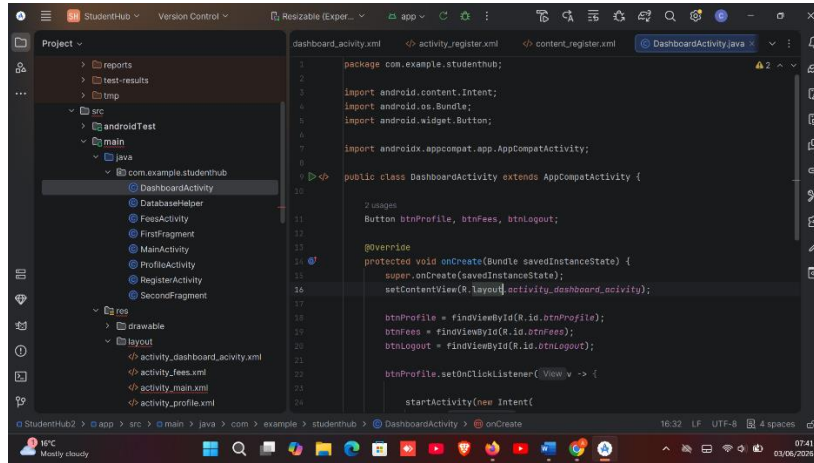
Objectives

- Implement SQLite Database
- Store user information
- Validate login credentials
- Create Profile screen
- Create Fees screen
- Test full application functionality

How i finished the built

I worked on storing user information and connecting the remaining screens in the application. I created a local database to save student account details such as usernames and passwords. When a user registers, their information is automatically stored in the database. I also linked the dashboard buttons to their respective screens. The Profile button displays the user's saved information, while the Fees button opens a screen showing the student's fee balance information.

DATABASEHELPER



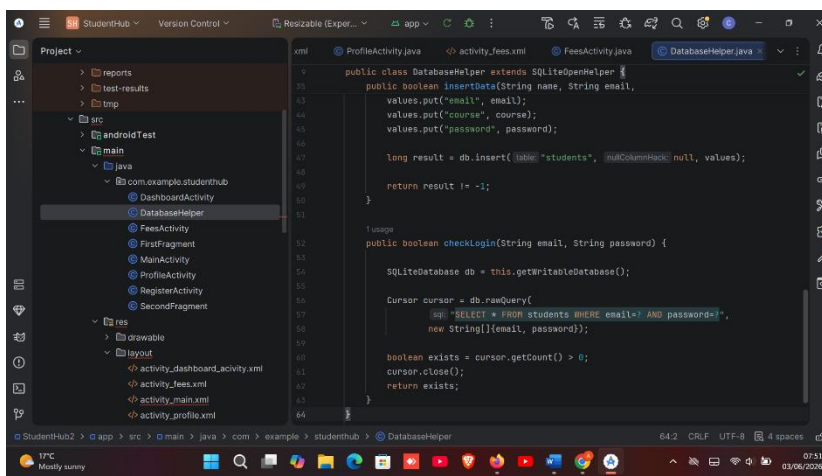
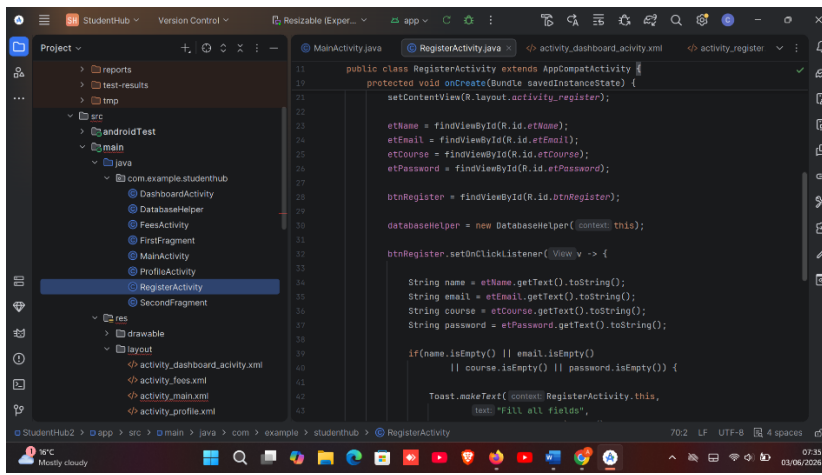
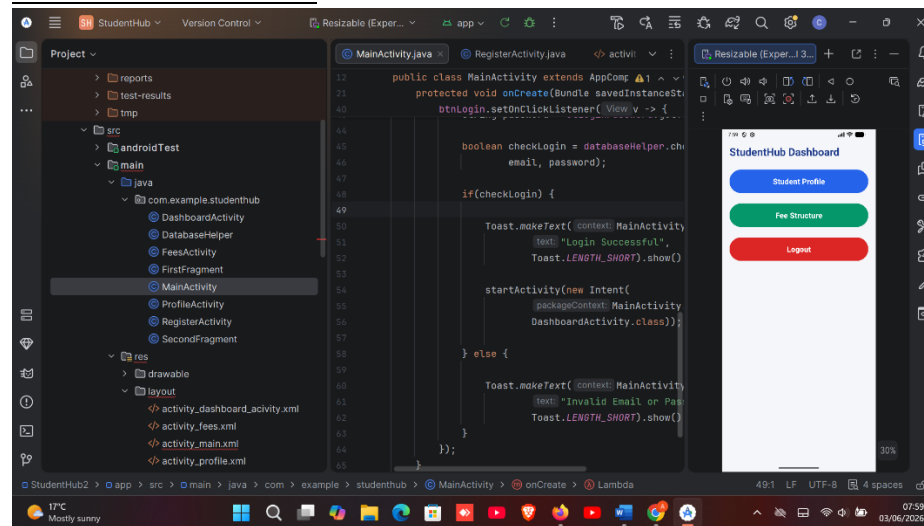


Figure 5.DASHBOARDHELPER

SUCCESSFUL LOGIN INPUT



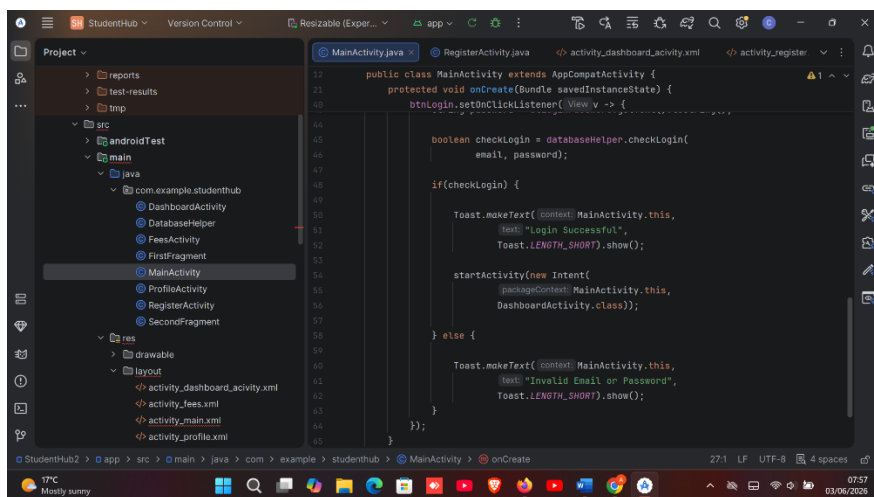
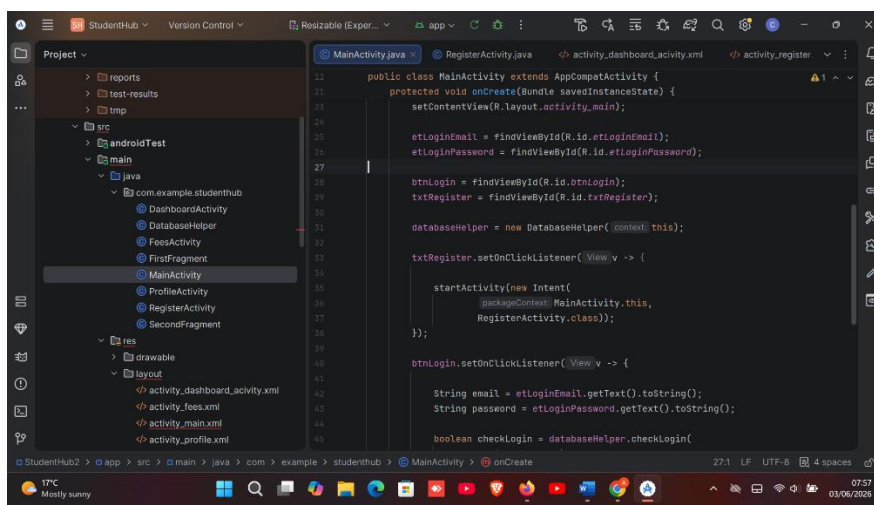
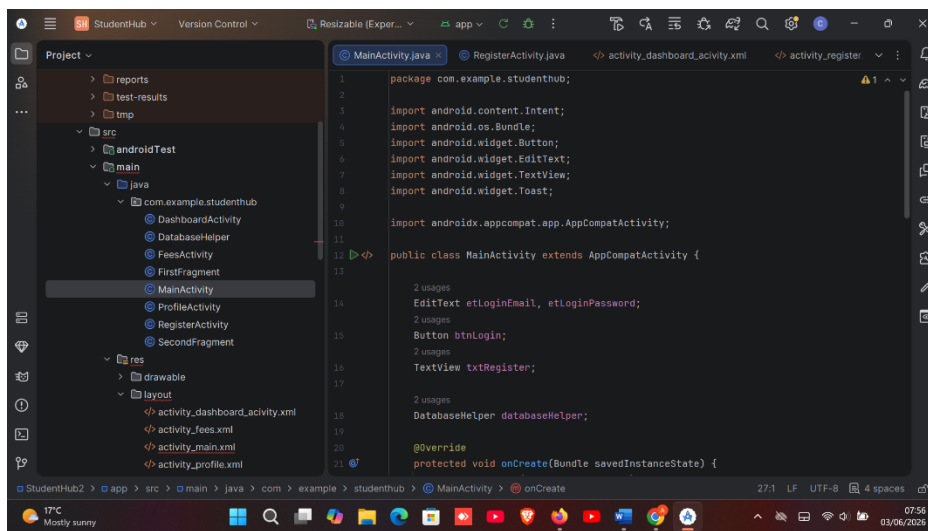


Figure 6LOG INPUTS AND CODE

REVIEW

PROFILE SCREEN

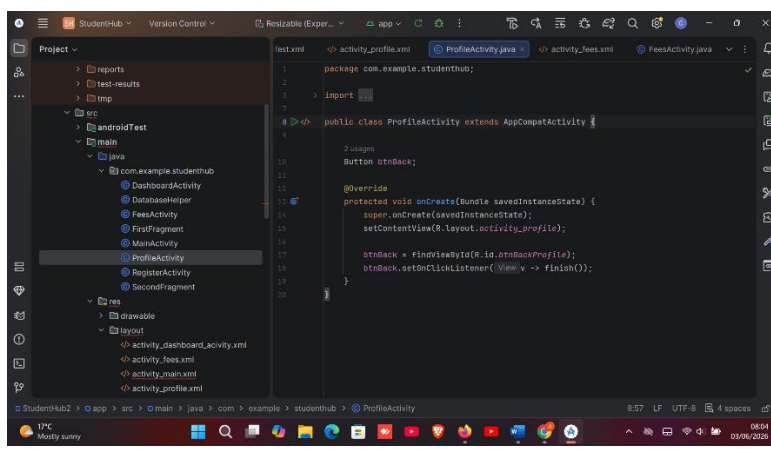
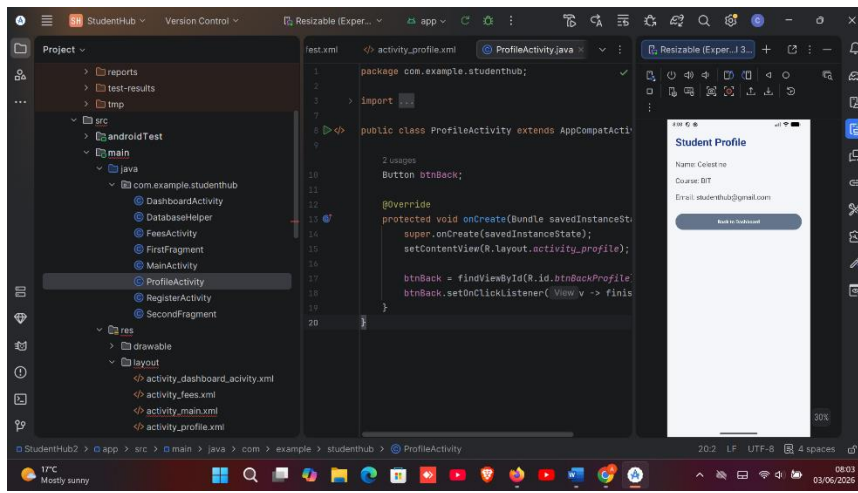
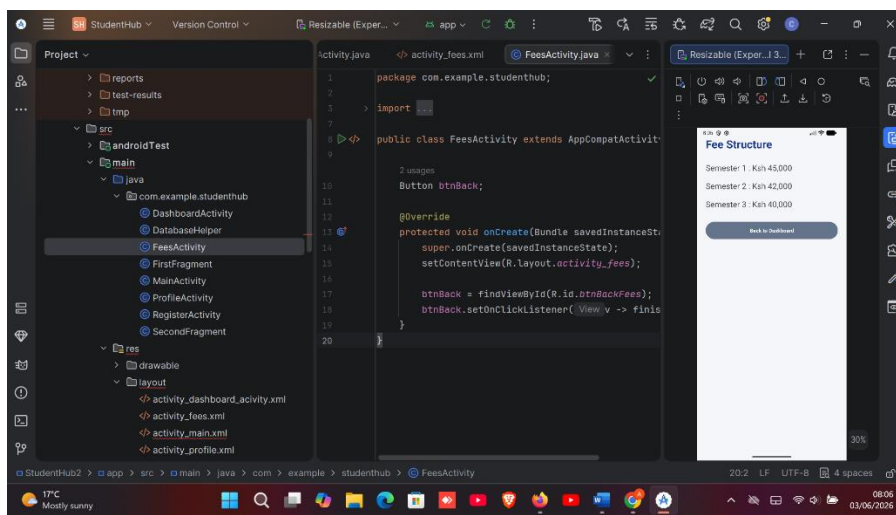


Figure 7PROFILE SCREENS AND CODE REVIEW

FEES SCREEN



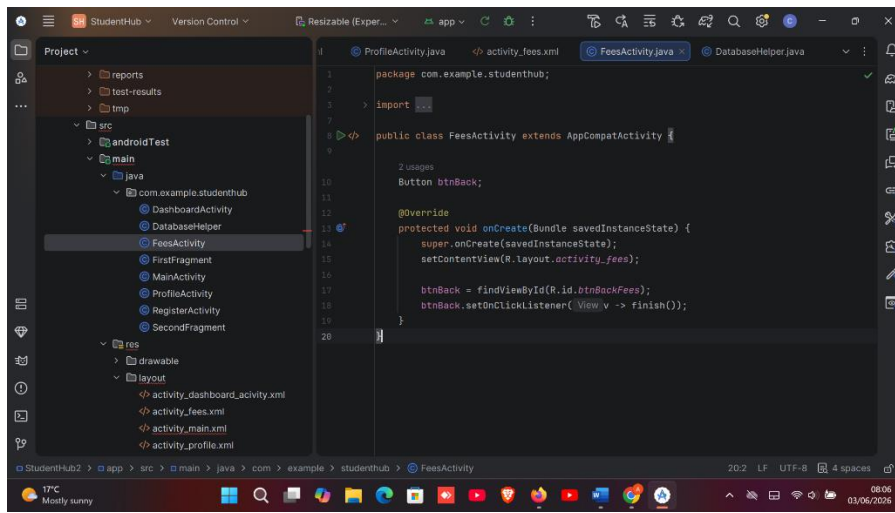
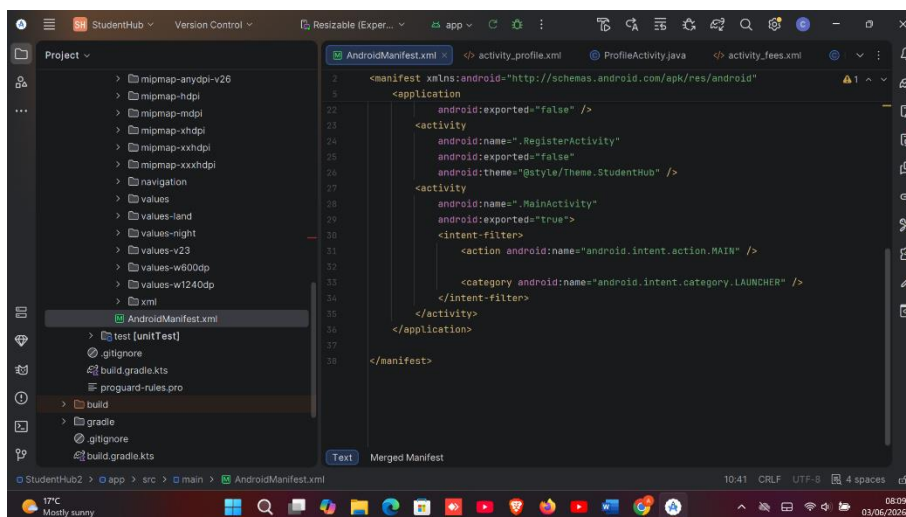
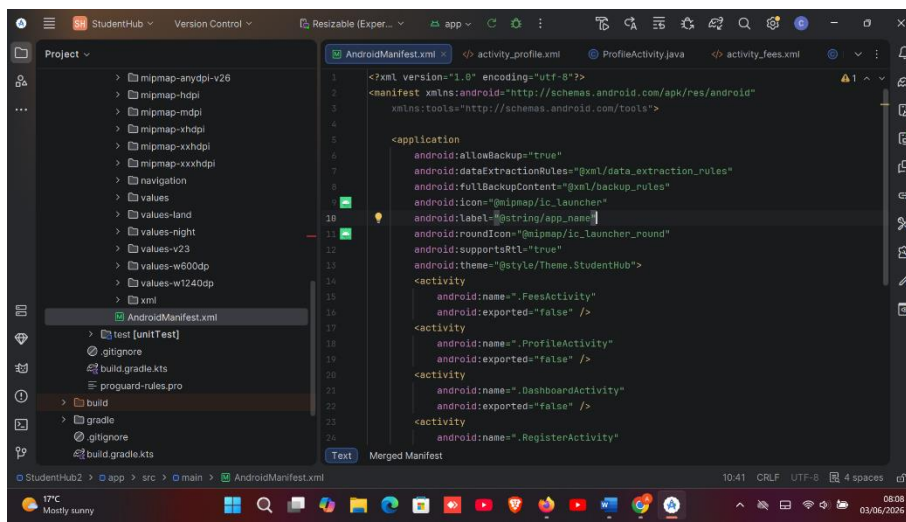


Figure 8FEES SCREEN AND

CODE REVIEW

ANDROIDMANIFEST.XML



Figure

9.ANDROIDMANIFEST.xml

CONCLUSION

The Student Hub mobile application was successfully developed using Android Studio, Java, XML, and SQLite Database technologies. The application achieved all intended objectives including user registration, login authentication, dashboard navigation, profile viewing, fee structure access, and logout functionality.

All system features were tested successfully using the Android Emulator, and the application operated correctly without errors.

